

Finding Triangles

Problem 955

A sequence $(a_n)_{n \geq 0}$ starts with $a_0 = 3$ and for each $n \geq 0$,

- if a_n is a **triangle number**, then $a_{n+1} = a_n + 1$;
- otherwise, $a_{n+1} = 2a_n - a_{n-1} + 1$.

The sequence begins:

3, 4, 6, 7, 9, 12, 16, 21, 22, 24, 27, 31, 36, 37, 39, 42, ...

where triangle numbers are marked red.

The 10th triangle number in the sequence is $a_{2964} = 1439056$.

Find the index n such that a_n is the 70th triangle number in the sequence.